

Pregame Pitcher Strikeout Projection - Resume Project Summary

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Repo work queue (not this resume): live approvals / next steps live in [docs/EXECUTION BACKLOG.md](#). This page is a portfolio summary only.

Primary project link: [GitHub repository](#).
Secondary long-form write-up: [Technical manuscript](#)

Goal

Estimate a starter’s strikeout rate before first pitch, project how many batters that starter will face, and convert the pair into expected strikeouts and $P(K \geq L)$ for common prop lines L —using only information available before first pitch:

$$k_{\text{rate}} \times \text{TBF} \rightarrow E[K] \rightarrow P(K \geq L)$$

Professional positioning

This project demonstrates a blended profile across:

- **AI/ML engineering:** end-to-end feature pipelines, freeze discipline, and reproducible deployment artifacts.
- **Applied statistics:** chronological validation, calibration diagnostics, and leakage-safe modeling controls.
- **Quant decisioning:** market-skill ranking vs book closes, policy-floor governance, and risk-aware deployment gates.

Models and headline metrics

Component	Model	Role
Strikeout rate (live)	Weighted 3-model LightGBM blend	Active production scorer (live_krate_ensemble.json)

Component	Model	Role
Batters faced (TBF)	Ridge (thin bullpen, 24 features)	Projected exposure
Counts / lines	Poisson on projected TBF (binomial one-line revert)	Expected K and $P(K \geq L)$
Line maps (live)	Per-line Platt (WS1c), shipped 2026-09-10	Replaces August isotonic pointer

Live policy (promoted 2026-09-11 from a 2025-lock selection): line floors + 4.5-over veto + 2.5/3.5 probation + morning edge cap $\emptyset.24$ + under-lean premium $+\emptyset.04$ on overs + DK+FD-only universe + offset clip $\pm\emptyset.02$ + 1/16-Kelly + postseason HOLD. A same-day robust-refusal arm was measured and reverted (2025 -0.2% vs base $+4.2\%$).

Policy-search diagnostics (pre-freeze $n=26$, 2026-07-30–08-17; historical): ROI $\emptyset.4363$, Sharpe $\emptyset.4438$. Audit 2026-09-01: **selection-window evidence**, not validated deployment edge. Superseded by the replay era below.

Universe close skill (live Poisson + WS1c, $n=19,533$): Brier $\emptyset.2204$ vs book $\emptyset.2162$ (skill $-\emptyset.0043$); ECE $\emptyset.021$ vs book $\emptyset.009$. Skill negative on all eight lines. Same starts **beat friend opens** (skill $+\emptyset.044$ at $\approx -12h$) and trail by morning.

Juiced replay (frozen probs at executable prices, rejects kept, $n=2,077$): flat-1u ROI **+7.3%** all-books / **+12.3%** DK+FD-only; 2025 $+4.2\%$ / 2026 $+12.6\%$ confirmatory. Fills unmodeled.

2025-lock champion (promoted with disclosed 2026 peek): 36-config family scored on 2025 only $\rightarrow$ floor $\emptyset.12$ / cap $\emptyset.24$ / under-lean / DK+FD: 2025 $n=558$ ROI **+15.5%** WR $\emptyset.60$ LCB **+8.0%**; White-lite $p<0.0005$; exclusions $\geq +12.9\%$; 2026 one-look repeat $n=485$ ROI **+15.6%**.

Paper money track (deduped 38 days, $n=323$): PnL $+\$576$ ($+2.0\%$ ROI), mean CLV $+\emptyset.70$ pp on $n_{clv}=195$; veto lane $+6.0\%$ ($n=263$) — paper prices, not fills. ≥ 50 real fills remain the money-truth gate.

Full-stack evaluation (production)

Evaluation gate	Result
Estimator tuning	Keep baseline LightGBM defaults; Ridge α tuned and persisted
Calibration + probability quality	Live WS1c Platt + Poisson; universe ECE $\emptyset.021$ / MCE $\emptyset.088$ on $n=19,533$ (book $\emptyset.009$ / $\emptyset.019$). Search-lane ECE $\emptyset.0639$ on $n=26$ is labeled, not live.

Evaluation gate	Result
Population policy	Metrics conditional on PA ≥ 9 (~3.5% of first pitchers excluded)
Exogenous-exit governance	Source-tagged anomaly overrides + training mask are shipped; current walk-forward A/B effect is neutral under low historical tag density

Current governance winners

High-signal results from current artifacts:

- Universe close-matched panel (`rescore_cal_report.json`): live config skill `-0.0043` Brier vs book; ships closed ~27% of the frozen-config gap.
- Open-universe deduped sweep winner
(`ensemble_sweep_ranked_ensemble_full_aug21_deduped.csv`) — **search lane, not live quality:**
- blend `0.05 sparse72 / 0.45 sparse72_monotone / 0.50 final58`
- ROI `0.6612`, Sharpe `0.9468`, Sortino `0.6954`, bets `35`
- Brier/LogLoss skill vs market: `+0.0825 / +0.0645`
- Active deployment king
(`open_top3_transfer_manual_replay_aug21_deduped_top3_from_dedupedsweep.json`):
- blend `0.00 sparse72 / 0.60 sparse72_monotone / 0.40 final58` (frozen 2026-08-21)
- **policy-search** ROI `0.4363`, PnL `1208.55`, Sharpe `0.4438`, Sortino `0.4277`, max DD `0.1905` on pre-freeze `n=26` (2026-07-30–08-17) — not post-freeze OOS
- Duplicate-ticket diagnostics explicitly tracked: `123` duplicate groups, `246` duplicate tickets in manual-set audits.

Resume-ready takeaway

Leakage-safe Statcast → rate × exposure stack with nested chronological selection, scaled into a quant-governed production system: frozen-model replay at executable prices, pre-registered 2025-lock policy selection (36 configs, White-checked, disclosed-peek judged), per-line Platt calibration, Poisson counts, and a live board governed by regression pins. Universe skill trails books on probabilities (`-0.0043`); the money comes from choice — clock, tickets, books — and the ledger proves each choice: veto, lean, cap, DK+FD-only. A same-day arm that didn't validate (robust-refusal, 2025 `-0.2%`) was reverted within hours with the evidence cited. No fills-gated money claim.

Portfolio claim: production-minded ML engineering under a strict information constraint, strengthened by statistical rigor and quant-style decision governance — including the discipline to revert your own morning shipment by afternoon.

Copy/Paste resume bullets

- Built a leakage-safe MLB forecasting pipeline over pitch-level Statcast data (Polars + Python, 340+ pytest guards), enforcing chronological splits and pregame-only features, then wired it into a governed live decision system that scores a daily board, logs a paper ledger, and pages on failure.
- Designed a two-stage prediction stack ($k\text{-rate} \times \text{projected TBF}$) with Poisson line probabilities and per-line Platt calibration; measured it against book closes at universe scale ($n=19,533$: Brier 0.2204 vs 0.2162) and at executable juiced prices with rejects retained (flat-1u $+7.3\%$ all-books / $+12.3\%$ DK+FD).
- Ran a pre-registered 36-config policy selection on 2025 data (slate-clustered LCB champion: floor 0.12 / cap 0.24 / under-lean / DK+FD-only, ROI $+15.5\%$), stress-checked it (White-lite $p<0.0005$, exclusions $\geq +12.9\%$), confirmed on a disclosed 2026 peek ($+15.6\%$), and promoted it to production behind regression pins — reverting a same-day arm the data rejected.
- Productionized execution hygiene: atomic ledger writes, settle hard-guards, shared API fetch, failure-banner alerting, and a reproducibility contract (raw JSON + frozen hashes + policy version reconstruct every ticket).

Interview framing (30 seconds)

I built a production-first sports forecasting system where the hard part was not model accuracy — books beat us on probabilities at every line — but turning a frozen model into executable edge: bet earlier (we beat the opener by $+0.044$), filter harder (a pre-registered 36-policy selection with White-checked significance picked floor/cap/lean/DK+FD-only for $+15.5\%$ on 2025, repeated $+15.6\%$ on 2026), and report honestly (rejected tickets kept, peeks disclosed, a same-day shipment reverted when selection-year data rejected it).

Code first: [GitHub repository](#) · Paper: [manuscript PDF](#)